



BUSINESS CASE

WAH FOR HOSPITALS (WAH4H)

**ASIA PACIFIC COLLEGE
3 HUMABON PLACE, MAGALLANES VILLAGE
MAKATI CITY, METRO MANILA 1232**

2026





1. Executive Summary

This Business Case documents the justification, scope, financial basis, and strategic alignment for the WAH for Hospitals (WAH4H) Hospital Information System project. It provides the project sponsor, academic evaluators, and key stakeholders with a structured, evidence-based rationale for approving and funding project execution.

1.1. Issue

The Wireless Access for Health (WAH) organization currently lacks a standardized, scalable, and regulation-compliant Hospital Information System. The existing codebase, developed during PBL1, is a monolithic, locally isolated application inaccessible for centralized demonstration, government evaluation, or operational deployment. It cannot support the clinical workflows required by Philippine healthcare standards (RA 10173, RA 11223, PhilHealth eClaims, PH Core FHIR R4).

Without immediate action, WAH risks failing to secure institutional buy-in from the government officials required for full-scale deployment. In its current state, the system delivers zero operational value.

1.2. Anticipated Outcomes

Upon successful completion, the WAH4H project will deliver: a fully containerized, demonstration-ready Hospital Information System covering 11 clinical modules (PBL1 + PBL2), deployed on WAH's Proxmox bare-metal hypervisor; full compliance with Philippine healthcare regulations (PH Core FHIR R4, PhilHealth eClaims, ICD-10-PH, LOINC, SNOMED CT, PSGC, MIMS PH, RVS, RA 10173, and RA 11223); a standardized 3-tier deployment pipeline (Local → Sandbox → Production) eliminating environment inconsistencies; a complete documentation suite: Training Manual, End-User Manual, Technical Manual, UAT Documentation, and Source Code Repository; and a platform enabling WAH to present to government officials and secure institutional commitment for full production deployment.

1.3. Recommendation



Team Pseudoers recommends immediate approval and execution of the WAH4H project. The recommended approach involves Docker containerization of the existing PBL1 codebase, deployment to WAH's Proxmox bare-metal hypervisor via isolated virtual machines secured through Cloudflare Tunnel, and iterative integration of PBL2 features using an Agile Scrum framework, all within a 9.4-week MVP schedule with zero actual financial liability to WAH.

1.4. Justification

This approach is justified on four grounds: it leverages already-allocated WAH hardware, eliminating capital expenditure; Docker containerization guarantees environment parity and eliminates deployment risk; a hybrid predictive-agile project management framework ensures both academic compliance and technical adaptability; and delivery of a demonstration-ready system is the critical prerequisite for WAH securing government support for operational scaling. The quantified benefit-cost profile demonstrates a strongly positive return even under conservative assumptions, with ₱97,632 per year in cloud hosting costs avoided, estimated 40% reduction in administrative processing time, and elimination of third-party training vendor costs.

2. Business Case Analysis Team

The following individuals are responsible for the analysis, preparation, and approval of this Business Case. The business case is developed by the project sponsor in collaboration with the project manager and key stakeholders, as established in sound project governance practice.

- **Project Manager:** Elijah Josh R. Quibin. Responsibilities: overall execution, schedule, budget, stakeholder communication, and all project documentation artifacts.
- **Lead Developer / DevOps:** Jhon Lloyd Nicolas. Responsibilities: system architecture, backend (Django 5.x), FHIR R4 compliance, PBL2 security, infrastructure configuration.
- **Quality Assurance:** Mariyah Vanna Monique C. Chavez. Responsibilities: QA validation, PHCore R4 alignment, requirements traceability, UAT oversight, documentation quality review.



- **Scrum Master / Agile Developer:** John Kenneth L. Jajurie. Responsibilities: Agile Scrum ceremonies, sprint facilitation, sprint planning, daily standups.
- **Project Sponsor:** Sir Pey Canlas. Responsibilities: executive authorization, business requirements, and final approval authority.
- **PROJMAN Professor:** Jose Eugenio Quesada. Responsibilities: academic oversight, standards alignment, and final evaluation.
- **Project Adviser:** Felino Calderon III. Responsibilities: technical guidance, architecture review, and feasibility evaluation.

The following table lists the Business Case Analysis Team, each with their role and responsibilities:

Name	Role	Responsibilities
Elijah Josh R. Quibin	Project Manager	Overall execution, schedule, budget, stakeholder communication, and all project documentation artifacts.
Jhon Lloyd Nicolas	Lead Developer / DevOps	System architecture, backend (Django 5.x), FHIR R4 compliance, PBL2 security, infrastructure configuration.
Mariyah Vanna Monique C. Chavez	Quality Assurance	QA validation, PHCore R4 alignment, requirements traceability, UAT oversight, documentation quality review.
John Kenneth L. Jajurie	Scrum Master / Agile Developer	Agile Scrum ceremonies, sprint facilitation, sprint planning, daily standups.



Name	Role	Responsibilities
Sir Pey Canlas	Project Sponsor	Executive authorization, business requirements, and final approval authority.
Jose Eugenio Quesada	PROJMAN Professor	Academic oversight, standards alignment, and final evaluation.
Felino Calderon III	Project Adviser	Technical guidance, architecture review, and feasibility evaluation.

3. Problem Definition

3.1. Problem Statement

The WAH4H system, developed during PBL1, exists strictly as a monolithic, locally isolated Django application with no containerization, no standardized deployment pipeline, and no production hosting environment. In its current state, the system cannot be accessed by WAH stakeholders, government evaluators, or academic reviewers from any centralized location; does not meet production-readiness requirements for PH Core FHIR R4 compliance, PhilHealth eClaims processing, or RA 10173 data privacy obligations; is susceptible to deployment discrepancies across the four development machines used by Team Pseudoers, creating environment parity failures that block concurrent PBL2 feature development; and has no backup/restore procedures, no audit logging, and no role-based access control, all of which are prerequisites for clinical operation under Philippine law.

Without resolving these issues within the 9.4-week MVP schedule, WAH4H will fail to deliver any demonstrable value, either academically or operationally.

3.2. Organizational Impact

The WAH4H project affects the following organizational dimensions: Tools & Technology: current state has no centralized hosting; monolithic local codebase; no containerization.



WAH4H delivers Proxmox hypervisor deployment, Docker containerization, and Cloudflare Tunnel security layer. Processes: current manual deployment per developer machine transitions to a standardized 3-tier pipeline (Local → Sandbox → Production) with automated container builds and formal UAT sign-off. Roles & Responsibilities: no defined WAH IT operational roles currently; Training Manual enables WAH staff to operate and train others while the Technical Manual enables Sir Robert Michael Martinez to manage infrastructure independently. Compliance: no FHIR R4 schema enforcement or RA 10173 audit trail currently; WAH4H delivers full PH Core FHIR R4 compliance, JWT + RBAC security, eClaims module, ICD-10-PH and LOINC integration. Stakeholder Visibility: no visibility for government officials currently; demo-ready system enables formal presentation to governors and mayors for institutional buy-in.

The following table summarizes the organizational impact across each dimension, comparing current state with WAH4H delivery:

Dimension	Current State	WAH4H Delivers
Tools & Technology	No centralized hosting; monolithic local codebase; no containerization.	Proxmox hypervisor deployment, Docker containerization, and Cloudflare Tunnel security layer.
Processes	Manual deployment per developer machine.	Standardized 3-tier pipeline (Local → Sandbox → Production) with automated container builds and formal UAT sign-off.
Roles & Responsibilities	No defined WAH IT operational roles.	Training Manual enables WAH staff to operate and train others; Technical Manual enables Sir Robert Michael Martinez to manage infrastructure independently.



Dimension	Current State	WAH4H Delivers
Compliance	No FHIR R4 schema enforcement or RA 10173 audit trail.	Full PH Core FHIR R4 compliance, JWT + RBAC security, eClaims module, ICD-10-PH and LOINC integration.
Stakeholder Visibility	No visibility for government officials.	Demo-ready system enables formal presentation to governors and mayors for institutional buy-in.

3.3. Technology Migration

The technology migration follows six phases: Phase I, Docker Containerization: package backend (Django 5.x), frontend (ReactJS), and PostgreSQL into Docker containers with Docker Compose; acceptance: all containers build with zero critical errors. Phase II, Proxmox Hypervisor Setup: install Proxmox VE, configure VM network bridging, storage pools, resource allocation; acceptance: Proxmox dashboard accessible with correct resource allocation (16GB RAM, 320GB, 8 CPUs). Phase III, Sandbox Deployment & Testing: deploy containerized application, validate 100% environment parity, execute integration testing; acceptance: all integration tests passing, no critical build errors. Phase IV, PBL2 Feature Integration: deploy Admin, Appointments, and eClaims modules with JWT security, RBAC, audit logging, and PHCore FHIR DB refactor; acceptance: PBL2 features pass UAT sign-off by Ms. Anna Katrina Yturralde; current status: initial UAT feedback received, UAT-requested changes in progress, final UAT/sign-off pending. Phase V, Production VM Deployment: migrate from Sandbox to Production VM, configure Cloudflare Tunnel, support final UAT/re-testing; acceptance: system operational, Cloudflare Tunnel active, final UAT/sign-off pending re-testing acceptance by Ms. Anna Katrina Yturralde. Phase VI, Documentation & Handover: deliver all three manuals, UAT docs, Source Code Repository, conduct runbook training; acceptance: all documentation accepted by Anna Katrina Yturralde and Sir Robert Michael Martinez; WAH staff independently operates system.

The following table summarizes the six technology migration phases, each with its description and acceptance criteria:



Phase	Description	Acceptance Criteria
I — Docker Containerization	Package backend (Django 5.x), frontend (ReactJS), and PostgreSQL into Docker containers with Docker Compose.	All containers build with zero critical errors.
II — Proxmox Hypervisor Setup	Install Proxmox VE, configure VM network bridging, storage pools, resource allocation.	Proxmox dashboard accessible with correct resource allocation (16GB RAM, 320GB, 8 CPUs).
III — Sandbox Deployment & Testing	Deploy containerized application, validate 100% environment parity, execute integration testing.	All integration tests passing, no critical build errors.
IV — PBL2 Feature Integration	Deploy Admin, Appointments, and eClaims modules with JWT security, RBAC, audit logging, and PHCore FHIR DB refactor.	PBL2 features pass UAT sign-off by Ms. Anna Katrina Yturralde; current status: initial UAT feedback received, UAT-requested changes in progress, final UAT/sign-off pending.
V — Production VM Deployment	Migrate from Sandbox to Production VM, configure Cloudflare Tunnel, support final UAT/re-testing.	System operational, Cloudflare Tunnel active, final UAT/sign-off pending re-testing acceptance by Ms. Anna Katrina Yturralde.
VI — Documentation & Handover	Deliver all three manuals, UAT docs, Source Code Repository, conduct runbook training.	All documentation accepted by Anna Katrina Yturralde and Sir Robert Michael Martinez; WAH staff



Phase	Description	Acceptance Criteria
		independently operates system.

4. Project Overview

4.1. Project Description

The WAH4H project will containerize the existing FHIR-compliant Hospital Information System codebase using Docker and deploy it to WAH's privately managed Proxmox Virtual Environment. Team Pseudoers employs a Hybrid Development Life Cycle: high-level planning follows a predictive framework for academic documentation compliance, while technical execution follows an iterative Agile (Scrum) framework enabling concurrent PBL2 feature development alongside deployment stabilization. The system covers the complete clinical workflow, from patient registration through discharge, billing, pharmacy, laboratory, and PhilHealth eClaims processing, across 11 modules. The final deliverable is a demonstration-ready system enabling WAH to secure government institutional commitment for full production deployment.

4.2. Goals and Objectives

All objectives follow the SMART framework and align with WAH's organizational strategy and APC BSCS-SS231 academic requirements. Standardize Deployment: containerize 100% of PBL1 modules via Docker by April 24, 2026, with zero critical Docker build errors across all environments (owner: Jhon Lloyd Nicolas). 3-Tier Test Environments: create 100% environment-parity Proxmox pipeline by May 8, 2026, so that Sandbox deployment achieves full parity and all integration tests pass (owner: Chavez, QA). PBL2 Feature Integration: integrate Admin, Appointments, and eClaims modules with JWT/RBAC security by May 15, 2026, where all PBL2 features must pass UAT sign-off by Ms. Anna Katrina Yturalde; current status: initial UAT feedback received, UAT-requested changes in progress, final UAT/sign-off pending (owner: Jhon Lloyd Nicolas). Regulatory Compliance: achieve 100% compliance with PH Core FHIR R4, PhilHealth eClaims, ICD-10-PH, RA 10173, and RA 11223 before UAT (owner: Chavez, QA). Government Demo-Readiness: deliver fully functional demo on WAH Proxmox infrastructure by May–June 2026, with the system



demonstrated to WAH management; formal sign-off from Ms. Anna Katrina Yturralde, then Sir Pey Canlas (owner: Quibin, PM). Documentation Delivery: complete Training, End-User, and Technical Manuals plus UAT docs and source code by project closure (owner: Quibin, PM).

The following table summarizes the SMART objectives, each with its target, date, and owner:

Objective	Target & Date	Owner
Standardize Deployment	Containerize 100% of PBL1 modules via Docker by April 24, 2026 — zero critical Docker build errors across all environments.	Jhon Lloyd Nicolas
3-Tier Test Environments	Create 100% environment-parity Proxmox pipeline by May 8, 2026 — Sandbox deployment achieves full parity; all integration tests pass.	Chavez, QA
PBL2 Feature Integration	Integrate Admin, Appointments, and eClaims modules with JWT/RBAC security by May 15, 2026 — all PBL2 features must pass UAT sign-off by Ms. Anna Katrina Yturralde; current status: initial UAT feedback received, UAT-requested changes in progress, final UAT/sign-off pending.	Jhon Lloyd Nicolas
Regulatory Compliance	Achieve 100% compliance with PH Core FHIR R4, PhilHealth eClaims, ICD-10-	Chavez, QA



Objective	Target & Date	Owner
	PH, RA 10173, and RA 11223 before UAT.	
Government Demo-Readiness	Deliver fully functional demo on WAH Proxmox infrastructure by May–June 2026 — system demonstrated to WAH management; formal sign-off from Ms. Anna Katrina Yturralde, then Sir Pey Canlas.	Quibin, PM
Documentation Delivery	Complete Training, End-User, and Technical Manuals plus UAT docs and source code by project closure.	Quibin, PM

4.3. Project Performance

Key Performance Indicators: Schedule: milestone completion vs. baseline schedule; target: all milestones on or before target dates; method: weekly sprint review and Gantt comparison. Quality (Docker): zero critical build errors in Sandbox and Production; method: Docker build logs and QA validation sign-off. Quality (UAT): all client UAT feedback items resolved before next UAT/sign-off cycle; method: UAT feedback log and next UAT/sign-off documentation accepted by Ms. Anna Katrina Yturralde. Compliance: 100% of required standards implemented and validated; method: compliance checklist and QA audit. Scope: all approved PBL2 features delivered; no unapproved additions; method: sprint backlog reviews and PM scope log. Stakeholder: WAH team can operate and train staff without team support; method: runbook training sign-off and Anna Katrina Yturralde acceptance. Budget: total labor ≤ ₱185,158.50 at role-based rates (Elijah ₱594/hr, Jhon ₱625/hr, Mariyah ₱363/hr, Jajurie ₱309/hr); method: timesheet tracking and PM budget log.



The following table summarizes the Key Performance Indicators, each with its target and measurement method:

KPI	Target	Method
Schedule	All milestones on or before target dates (milestone completion vs. baseline schedule).	Weekly sprint review and Gantt comparison.
Quality (Docker)	Zero critical build errors in Sandbox and Production.	Docker build logs and QA validation sign-off.
Quality (UAT)	All client UAT feedback items resolved before next UAT/sign-off cycle.	UAT feedback log and next UAT/sign-off documentation accepted by Ms. Anna Katrina Yturralde.
Compliance	100% of required standards implemented and validated.	Compliance checklist and QA audit.
Scope	All approved PBL2 features delivered; no unapproved additions.	Sprint backlog reviews and PM scope log.
Stakeholder	WAH team can operate and train staff without team support.	Runbook training sign-off and Anna Katrina Yturralde acceptance.
Budget	Total labor ≤ ₱185,158.50 at role-based rates (Elijah ₱594/hr, Jhon ₱625/hr, Mariyah ₱363/hr, Jajurie ₱309/hr).	Timesheet tracking and PM budget log.

4.4. Project Assumptions



A1: WAH Proxmox server (16GB RAM, 320GB, 8 CPUs) is provisioned and accessible by Sir Robert Michael Martinez before the deployment phase (target: April 10, 2026); validation: physical server access confirmed by PM at Milestone 2. A2: WAH provides timely feedback within 48 hours on UAT results and documentation reviews; validation: communication log tracked by PM; escalation to Sir Pey Canlas if breached. A3: Cloudflare service (\$20/month, existing) remains active throughout the project period; validation: confirmed monthly by PM. A4: WAH independently completes PhilHealth accreditation and NHFR registration prior to eClaims go-live; validation: prerequisite checklist confirmed before eClaims UAT phase. A5: All four team members maintain consistent physical or remote access to the Proxmox server during deployment phases; validation: VPN access confirmed during Sandbox phase. A6: WAH4P/WAH4PC interoperability specification remains in-scope but dependency-gated pending WAH technical specification from Sir Robert Michael Martinez; validation: specification document received and reviewed by QA before Milestone 9. A7: No new PBL2 features added beyond approved scope after May 8, 2026 scope freeze; validation: change request log; sponsor approval required for any additions.

The following table summarizes the project assumptions, each with its validation approach:

ID	Assumption	Validation
A1	WAH Proxmox server (16GB RAM, 320GB, 8 CPUs) is provisioned and accessible by Sir Robert Michael Martinez before the deployment phase (target: April 10, 2026).	Physical server access confirmed by PM at Milestone 2.
A2	WAH provides timely feedback within 48 hours on UAT results and documentation reviews.	Communication log tracked by PM; escalation to Sir Pey Canlas if breached.
A3	Cloudflare service (\$20/month, existing) remains active throughout the project period.	Confirmed monthly by PM.



ID	Assumption	Validation
A4	WAH independently completes PhilHealth accreditation and NHFR registration prior to eClaims go-live.	Prerequisite checklist confirmed before eClaims UAT phase.
A5	All four team members maintain consistent physical or remote access to the Proxmox server during deployment phases.	VPN access confirmed during Sandbox phase.
A6	WAH4P/WAH4PC interoperability specification remains in-scope but dependency-gated pending WAH technical specification from Sir Robert Michael Martinez.	Specification document received and reviewed by QA before Milestone 9.
A7	No new PBL2 features added beyond approved scope after May 8, 2026 scope freeze.	Change request log; sponsor approval required for any additions.

4.5. Project Constraints

Time: 9.4-Week MVP Schedule, which requires the project to complete March 23 – May 27, 2026. Budget: ₱0 WAH Financial Liability, meaning WAH bears no actual financial cost; simulated labor budget capped at ₱185,158.50 at role-based rates (Elijah ₱594/hr, Jhon ₱625/hr, Mariyah ₱363/hr, Jajurie ₱309/hr). Scope: No External SSL / Cloud Routing, which is strictly out of scope. Scope: No Post-Delivery Maintenance, so bug fixes and enhancements after formal turnover are not included. Resources: 4-Member Fixed Team, so no additional developers can be added mid-project. Technical: WAH Proxmox Hardware Limits, meaning the system must be deployable on WAH's existing server spec (16GB RAM,



320GB, 8 CPUs). Regulatory: PhilHealth / NHFR Prerequisites, since PhilHealth accreditation and NHFR registration are WAH prerequisites; team cannot unblock these.

The following table summarizes the project constraints by category:

Category	Constraint	Detail
Time	9.4-Week MVP Schedule	Project must complete March 23 – May 27, 2026.
Budget	₱0 WAH Financial Liability	WAH bears no actual financial cost; simulated labor budget capped at ₱185,158.50 at role-based rates (Elijah ₱594/hr, Jhon ₱625/hr, Mariyah ₱363/hr, Jajurie ₱309/hr).
Scope	No External SSL / Cloud Routing	Strictly out of scope.
Scope	No Post-Delivery Maintenance	Bug fixes and enhancements after formal turnover not included.
Resources	4-Member Fixed Team	No additional developers can be added mid-project.
Technical	WAH Proxmox Hardware Limits	System must be deployable on WAH's existing server spec (16GB RAM, 320GB, 8 CPUs).
Regulatory	PhilHealth / NHFR Prerequisites	PhilHealth accreditation and NHFR registration are WAH prerequisites; team cannot unblock these.



4.6. Major Project Milestones

Milestones are aligned with the Project Charter (v1.2) and tracked against the Schedule Management Plan. Status reflects current state as of May 2026.

No.	Milestone	Date	Status	Phase
1	Project Plan, Charter & SOW: Drafted and Approved	03/24/2026	DONE	Initiation
2	Proxmox Hypervisor Installation & VM Network Bridging	04/10/2026	DONE	Planning
3	System Architecture & Docker Containerization Design	04/24/2026	DONE	Design
4	Sandbox Integration Testing (100% Environment Parity)	05/08/2026	DONE	Execution
5	PBL2 Feature Integration (Admin, Appointments, eClaims, Security)	05/15/2026	IN PROGRESS	Execution
6	Final Deployment:	05/22/2026	PENDING	Execution



No.	Milestone	Date	Status	Phase
	Proxmox Production Initialization			
7	Client UAT Feedback Received: UAT- requested development changes pending; next UAT/sign- off cycle pending	05/18/2026	IN PROGRESS	Execution
8	Documentation Completion: All 3 Manuals + UAT Docs	TBD	IN PROGRESS	Execution
9	WAH4P/WAH4PC Integration & Data Sync Validation (dependency- gated pending WAH technical specification)	TBD	PENDING	Execution
10	Formal Turnover, Runbook Training & Client Handoff	May–Jun 2026	AT RISK	Closure
11	Formal Project Closure: Ms. Anna Katrina Yturralde	May–Jun 2026	AT RISK	Closure



No.	Milestone	Date	Status	Phase
	approval, then Sir Pey Canlas final sign-off			

**TBD milestones require internal team alignment. May–June 2026 deadline is AT RISK — early communication to Sir Pey Canlas required if slipping.*

5. Strategic Alignment

WAH Digital Health Infrastructure, to establish a scalable, sovereign, locally managed digital health platform: WAH4H deploys a 100% locally managed HIS on WAH's Proxmox server, eliminating cloud dependency and ensuring data sovereignty. WAH Regulatory Compliance, to achieve full compliance with RA 10173, RA 11223, PhilHealth eClaims, FHIR R4: all 11 modules are built to PH Core FHIR R4 standards with PhilHealth eClaims, ICD-10-PH, LOINC, and SNOMED CT integration. WAH Government Partnership, to secure institutional buy-in from LGU officials: the demo-ready system is the primary vehicle for WAH's government presentations. WAH Operational Efficiency, to standardize clinical workflows from registration through discharge: WAH4H covers the complete clinical workflow in a single integrated platform. APC Academic Excellence (BSCS-SS231), to demonstrate professional-grade project management and software engineering competency: project is managed under a hybrid framework with full documentation artifacts. APC Industry-Relevant Delivery, to produce deliverables that meet real-world client standards: WAH4H is not a mock project; it is a real system for a real client (WAH) addressing a real healthcare gap in the Philippines.

The following table summarizes strategic alignment, mapping each strategic objective to how WAH4H delivers it:

Strategic Objective	Goal	How WAH4H Delivers
WAH Digital Health Infrastructure	Establish scalable, sovereign, locally managed digital health platform.	Deploys a 100% locally managed HIS on WAH's Proxmox server, eliminating



Strategic Objective	Goal	How WAH4H Delivers
		cloud dependency and ensuring data sovereignty.
WAH Regulatory Compliance	Achieve full compliance with RA 10173, RA 11223, PhilHealth eClaims, FHIR R4.	All 11 modules are built to PH Core FHIR R4 standards with PhilHealth eClaims, ICD-10-PH, LOINC, and SNOMED CT integration.
WAH Government Partnership	Secure institutional buy-in from LGU officials.	The demo-ready system is the primary vehicle for WAH's government presentations.
WAH Operational Efficiency	Standardize clinical workflows from registration through discharge.	Covers the complete clinical workflow in a single integrated platform.
APC Academic Excellence (BSCS-SS231)	Demonstrate professional-grade project management and software engineering competency.	Project is managed under a hybrid framework with full documentation artifacts.
APC Industry-Relevant Delivery	Produce deliverables that meet real-world client standards.	WAH4H is not a mock project — it is a real system for a real client (WAH) addressing a real healthcare gap in the Philippines.

6. Cost Benefit Analysis

The WAH4H project operates under a unique financial structure: WAH assumes ₱0 liability for Team Pseudoers' simulated labor and owns all post-closure infrastructure costs.

Conversion rate: \$1 ≈ ₱56.50 (May 2026). One-Time Labor: Team Pseudoers Simulated Labor at role-based rates (Elijah ₱594/hr, Jhon ₱625/hr, Mariyah ₱363/hr, Jajurie ₱309/hr);



EVM Labor BAC ₱185,158.50; Total Authorized Budget ₱252,741.35, a Team liability. One-Time Infrastructure: Proxmox Server Setup (WAH-owned bare-metal, no procurement required), at ₱0 WAH liability. Recurring Infrastructure: Cloudflare Tunnel + DDoS Protection (\$20/month × ₱56.50 × 12), at ₱13,560/yr WAH liability. Cost Avoided: Equivalent Linode Cloud Hosting (\$144/month × ₱56.50 × 12) eliminated by Proxmox deployment, with ₱97,632/yr SAVED.

The following table summarizes the cost structure, including liability allocation:

Cost Category	Item	Amount	Liability
One-Time Labor	Team Pseudoers Simulated Labor at role-based rates (Elijah ₱594/hr, Jhon ₱625/hr, Mariyah ₱363/hr, Jajurie ₱309/hr); EVM Labor BAC ₱185,158.50	Total Authorized Budget ₱252,741.35	Team liability
One-Time Infrastructure	Proxmox Server Setup (WAH-owned bare-metal, no procurement required)	₱0	₱0 WAH liability
Recurring Infrastructure	Cloudflare Tunnel + DDoS Protection (\$20/month × ₱56.50 × 12)	₱13,560/yr	WAH liability
Cost Avoided	Equivalent Linode Cloud Hosting (\$144/month × ₱56.50 × 12) eliminated by Proxmox deployment	₱97,632/yr	SAVED



Tangible Benefits: Cloud Cost Avoidance ₱97,632/year saved; Clinical Workflow Efficiency ~40% admin time reduction; PhilHealth eClaims Enablement: faster reimbursement cycles; Staff Training Cost Reduction: ₱0 external training cost. Intangible Benefits: Government Institutional Buy-In: critical for project scaling; Data Sovereignty: regulatory risk mitigation; Healthcare Access Improvement: social impact; Academic Credibility & Portfolio: institutional reputation.

The following table summarizes the tangible and intangible benefits:

Type	Benefit	Detail
Tangible	Cloud Cost Avoidance	₱97,632/year saved.
Tangible	Clinical Workflow Efficiency	~40% admin time reduction.
Tangible	PhilHealth eClaims Enablement	Faster reimbursement cycles.
Tangible	Staff Training Cost Reduction	₱0 external training cost.
Intangible	Government Institutional Buy-In	Critical for project scaling.
Intangible	Data Sovereignty	Regulatory risk mitigation.
Intangible	Healthcare Access Improvement	Social impact.
Intangible	Academic Credibility & Portfolio	Institutional reputation.

7. Alternatives Analysis

Status Quo (No Project): Maintain WAH4H as isolated local PBL1 build. NOT SELECTED: system undeployable; zero value to WAH; no government demo possible; fails all project objectives. Managed Cloud Deployment (Linode/AWS): deploy to public cloud. NOT SELECTED: violates zero-cost constraint (₱97,632/yr recurring); creates data sovereignty risk under RA 10173; external vendor dependency. Docker on New Hardware: purchase new dedicated server. NOT SELECTED: capital expenditure violates ₱0 client liability constraint;



procurement timeline incompatible with 9.4-week schedule; WAH already owns suitable hardware. Monolithic Deployment Without Docker: deploy PBL1 directly without containerization. NOT SELECTED: creates deployment inconsistency; eliminates modular PBL2 integration; no rollback capability; fails environment parity requirement.

RECOMMENDED, Docker on WAH Proxmox: containerize existing codebase; deploy to WAH's Proxmox server via Docker Compose; secure via Cloudflare Tunnel. SELECTED: ₱0 recurring infrastructure cost; data sovereignty; environment parity; modular PBL2 integration; Cloudflare security; bare-metal competency demonstrated.

The following table summarizes the alternatives considered, each with its decision and rationale:

Alternative	Approach	Decision & Rationale
Status Quo (No Project)	Maintain WAH4H as isolated local PBL1 build.	NOT SELECTED: system undeployable; zero value to WAH; no government demo possible; fails all project objectives.
Managed Cloud Deployment (Linode/AWS)	Deploy to public cloud.	NOT SELECTED: violates zero-cost constraint (₱97,632/yr recurring); creates data sovereignty risk under RA 10173; external vendor dependency.
Docker on New Hardware	Purchase new dedicated server.	NOT SELECTED: capital expenditure violates ₱0 client liability constraint; procurement timeline incompatible with 9.4-week schedule; WAH already owns suitable hardware.
Monolithic Deployment Without Docker	Deploy PBL1 directly without containerization.	NOT SELECTED: creates deployment inconsistency;



Alternative	Approach	Decision & Rationale
		eliminates modular PBL2 integration; no rollback capability; fails environment parity requirement.
Docker on WAH Proxmox	Containerize existing codebase; deploy to WAH's Proxmox server via Docker Compose; secure via Cloudflare Tunnel.	RECOMMENDED — SELECTED: ₱0 recurring infrastructure cost; data sovereignty; environment parity; modular PBL2 integration; Cloudflare security; bare-metal competency demonstrated.





Approved by the Project Sponsor:

Date: _____

Sir Pey Canlas

Executive Director, Wireless Access for Health, Inc. (WAH)

